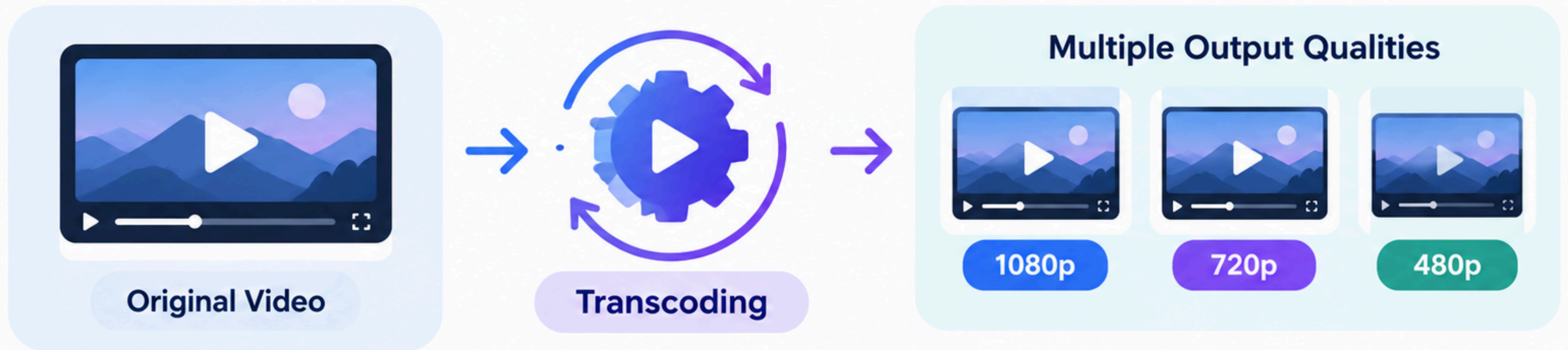


VIDEO TRANSCODING WEBSITE

Convert • **Optimize** • **Deliver**

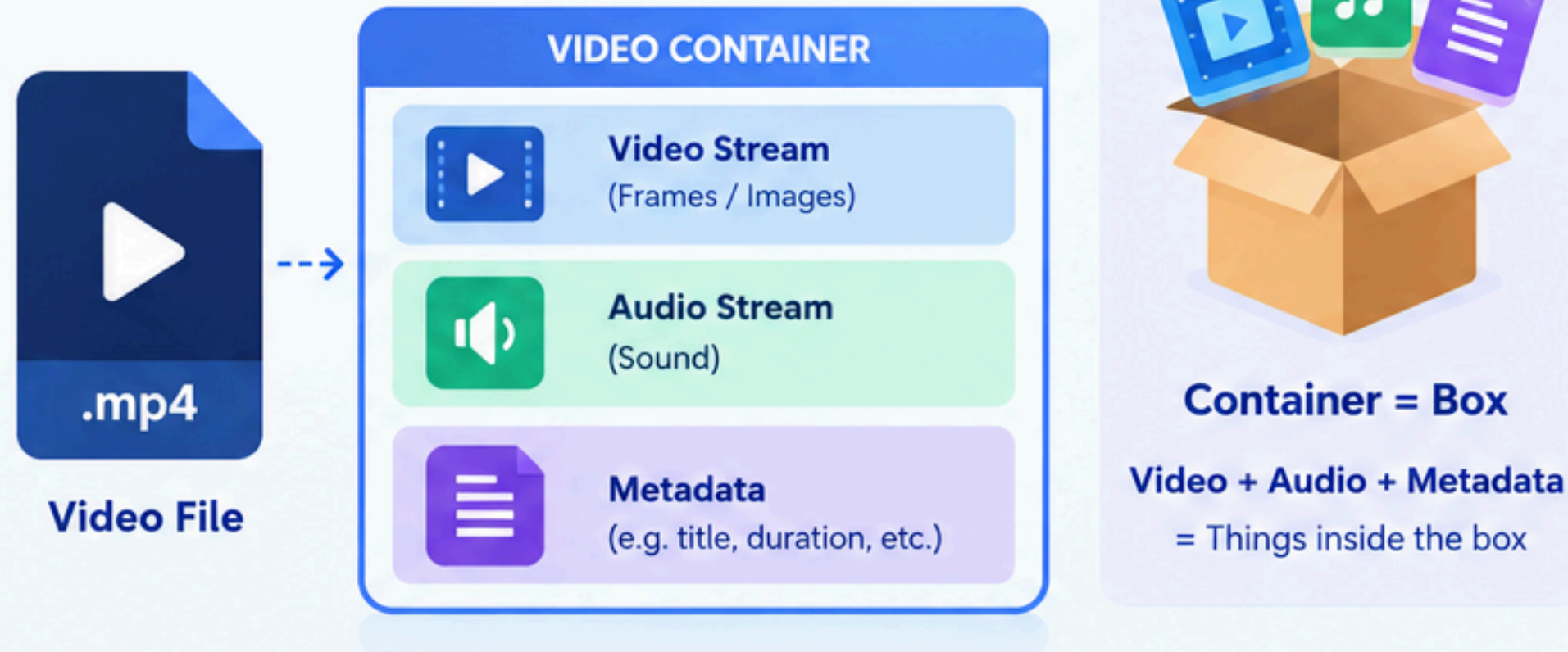
A Web-Based Video Processing Platform



What is Video Transcoding?

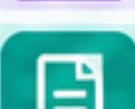
1. What is a Video?

A video is **more than just a file**. It contains multiple streams and information, all stored inside a container.



2. What Defines Video Quality?

A video has several important parameters:

	Resolution	Number of pixels in the video
	Bitrate	Amount of data used per second
	FPS	Number of frames per second
	Video Codec	Method to encode/decode video data
	Audio Codec	Method to encode/decode audio data
	Container	Format to store video/audio streams

Example

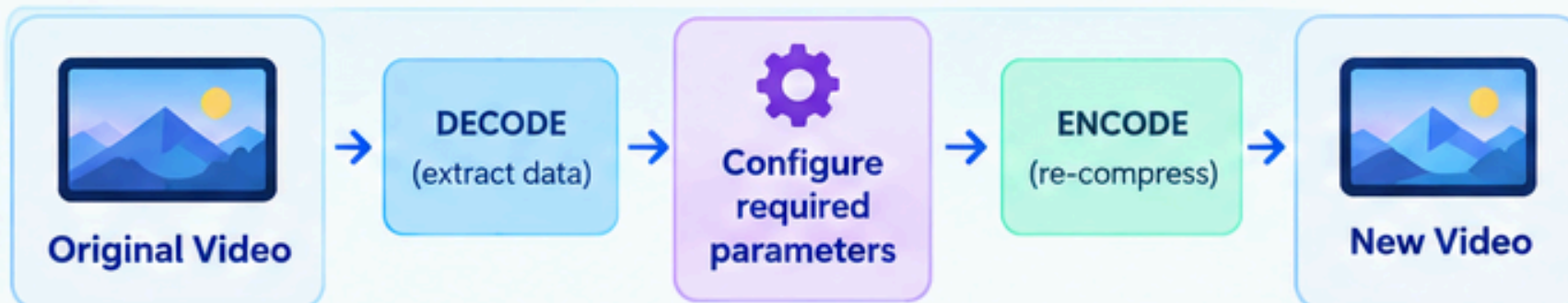


Resolution : 1920 × 1080
Codec : H.264
Bitrate : 8 Mbps
FPS : 30
Audio : AAC
Container : MP4

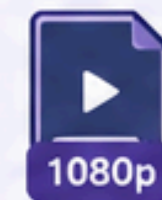
3. What is Video Transcoding?



Video transcoding is the process of decoding an existing video and **re-encoding** it into a new output configuration.



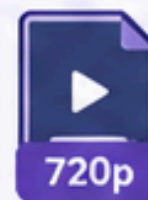
Before



Resolution: 1920 × 1080
Codec: H.264
Bitrate: 8 Mbps
FPS: 30



After



Resolution: 1280 × 720
Codec: H.264
Bitrate: 3 Mbps
FPS: 30

WHY DO WE NEED VIDEO TRANSCODING?

The Problem



Today, videos are increasingly recorded in high resolution.

For example:

4K / 1080p video



Higher amount of data



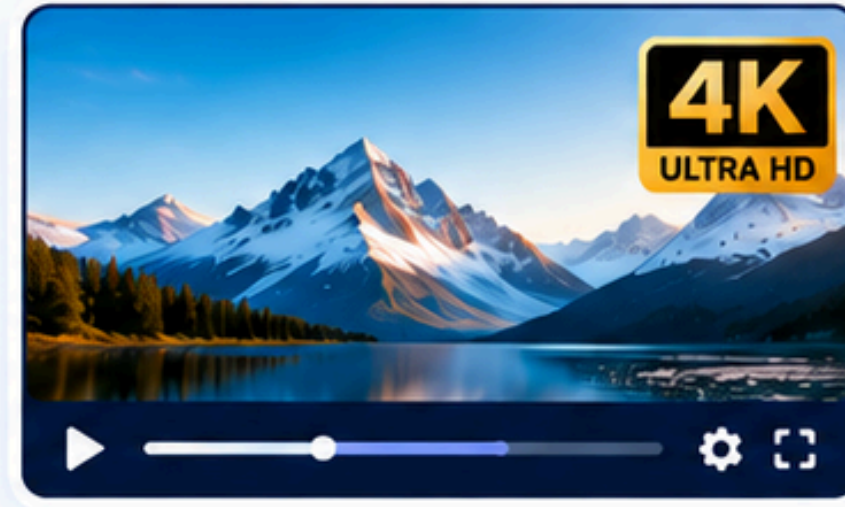
Larger file size



More storage required



More bandwidth required



Video Parameters



Resolution

(e.g. 4K, 1080p, 720p, 480p)



Codec

(e.g. H.264, H.265)



Bitrate

(e.g. 8 Mbps, 3 Mbps)



FPS

(e.g. 60, 30)



Audio

(e.g. AAC)

But every user has different requirements



Mobile user

Lower resolution
Smaller size
Less bitrate



Laptop user

Balanced quality
Medium bitrate
Good FPS



TV user

Higher resolution
Better quality
Higher bitrate



Slow internet

Lower resolution
Smaller bitrate
Lower FPS



Fast internet

Higher resolution
Higher bitrate
Better quality



Limited storage

Smaller file size
Lower bitrate
Optimized codec



One video configuration may not be suitable for every user, device and network condition.



WHERE IS VIDEO TRANSCODING USED?

Real-World Applications





YouTube — A Real-World Example

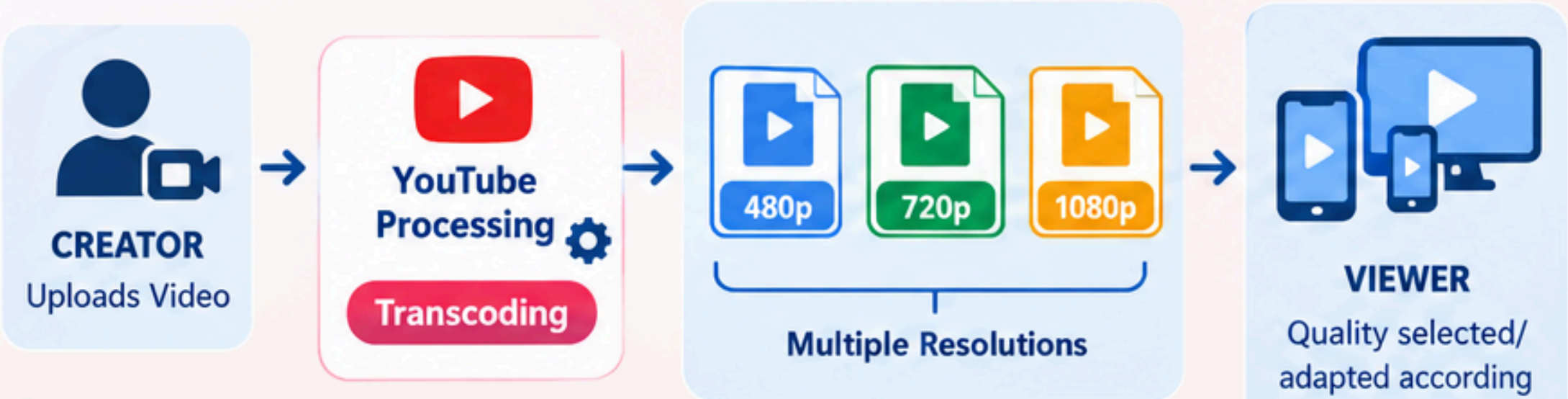
When a creator uploads a video to YouTube, the platform processes the video and makes it available in different resolutions:



YouTube’s documentation explains that video quality can adapt based on internet connection, screen size, browser and the available version of the video.

For **YouTube Live**, YouTube automatically transcodes the incoming stream into multiple output formats so viewers across different devices and network conditions can watch the stream.

How It Works (Conceptually)



Other Real-World Applications



Social Media
Different devices, resolutions and network conditions.



OTT Platforms
Multiple quality versions for different devices.



Online Education
Manageable video delivery over different networks.



Cloud Video Platforms
Video processing, storage and delivery.



Video Conferencing



OUR FRONTEND ARCHITECTURE

Modern Tools • Smooth User Experience • Seamless Integration



Technologies Used



React

User interface



React Router

Navigation between different pages



Axios

Communication with backend APIs



ReactPlayer

Video playback

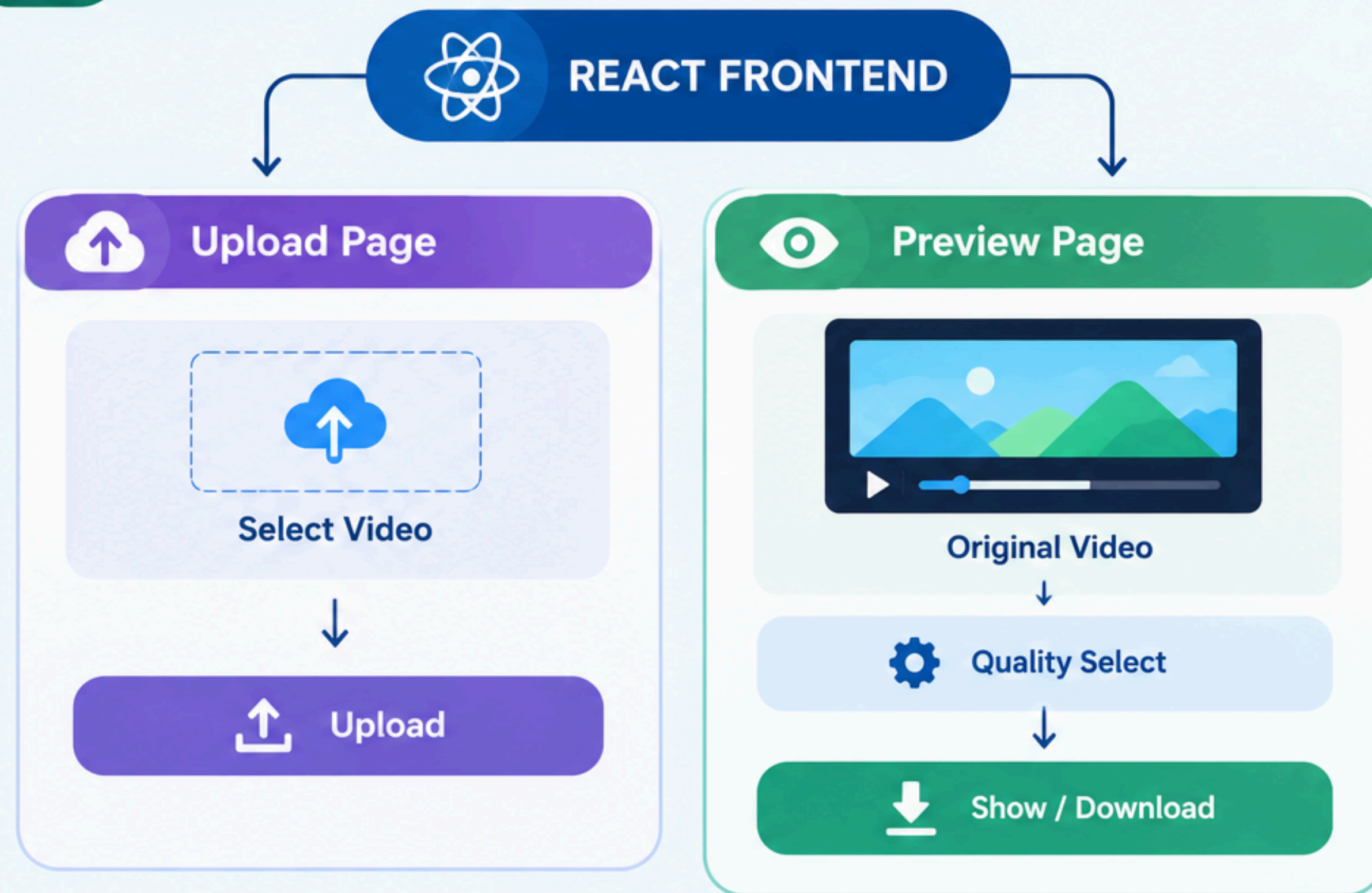


Tailwind CSS

UI styling

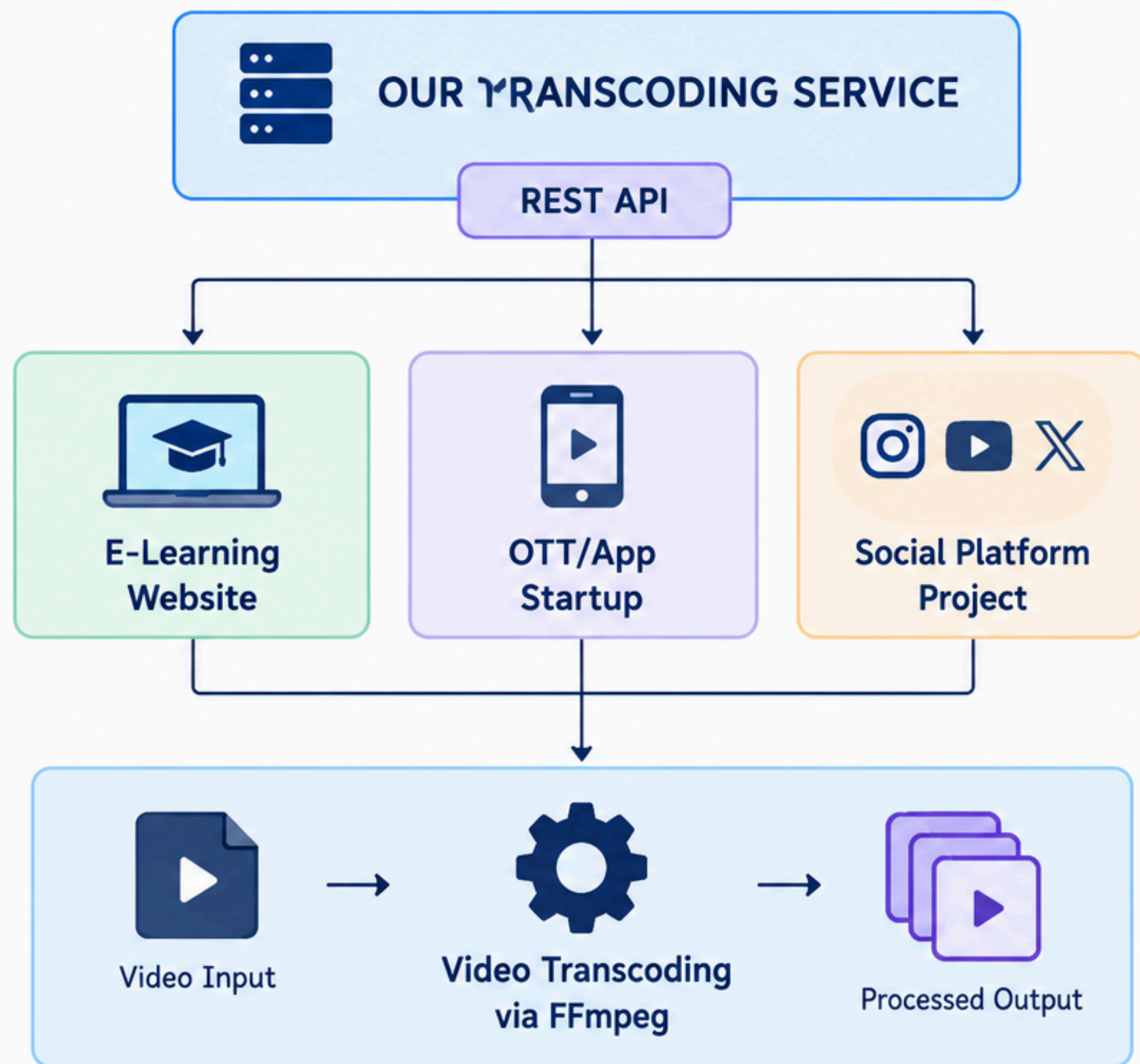


Architecture



Reusable Video Transcoding Service

Instead of building video-processing functionality from scratch every time, our transcoding system can work as a reusable backend service / API.



Why is this useful?

1



Reusability

The same transcoding service can be used by multiple applications instead of implementing video processing separately in every project.

2



API-Based Integration

A future application could simply send a video and its required configuration through an API and receive the processed output.

3



Faster Development

A startup or development team building a product that requires video processing would not have to develop the complete transcoding logic from scratch.

4



Microservice Architecture

Video processing can be kept as a separate service/microservice, independent from the main application.

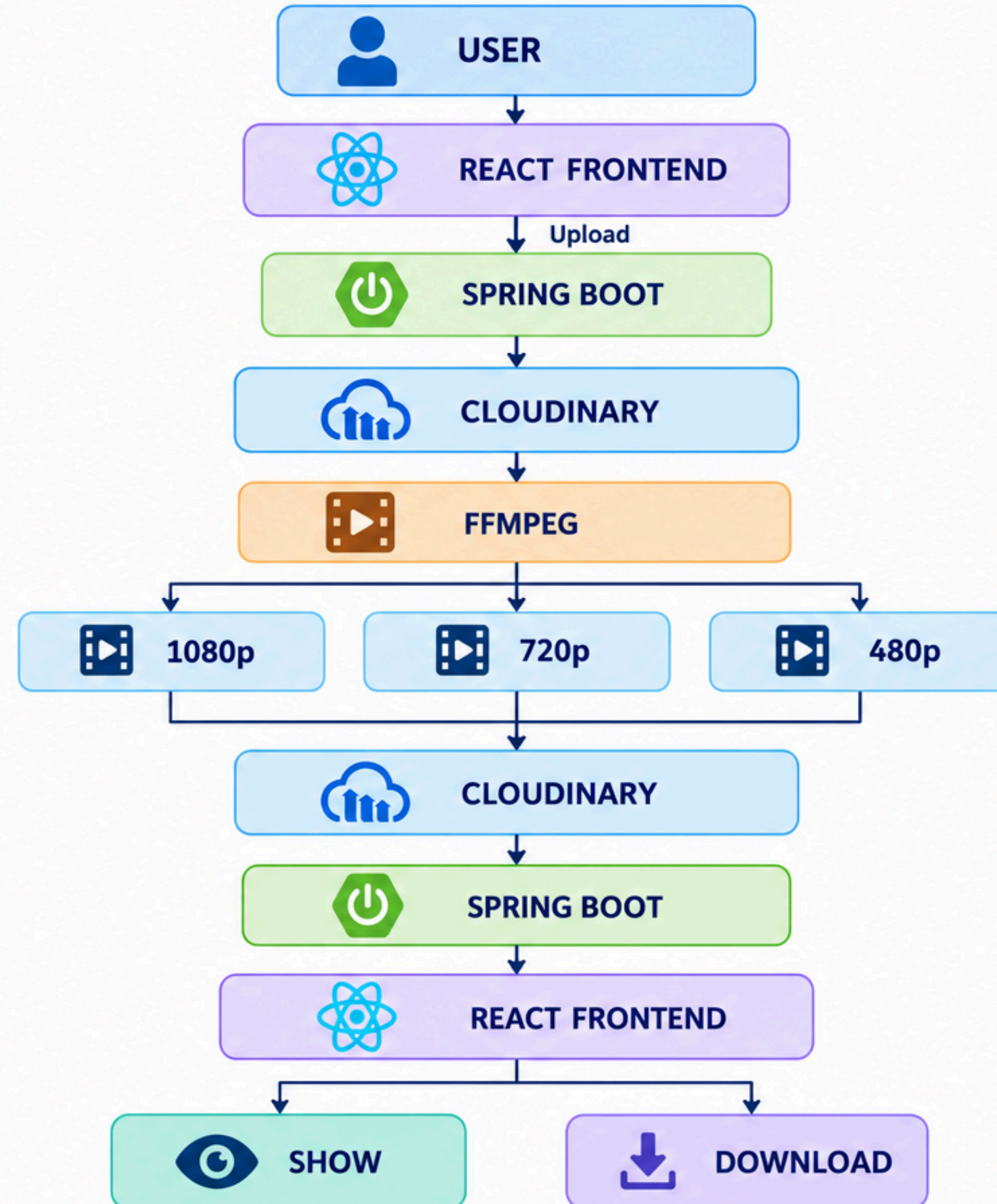
5



Easy to Extend

More output configurations, codecs, resolutions and processing options can be added to the service as the project evolves.

Complete System Architecture



THANK YOU



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